

Literature Status: the Column–Row Question for Complete Pick Spaces

Run fa_banach_001, lane 3 — 9 August 2026

Status. `literature_already_answered` (complete answer, with optimal constant).

Source question. In Clouâtre–Hartz, arXiv:1909.12883v2, page 4, immediately after the definition of the column–row property, the authors state:

It is an open question whether every normalized complete Nevanlinna–Pick space satisfies the column–row property with a finite constant.

The same paper’s Question 1.1 (page 2) asks whether

$$\text{Mult}(\mathcal{H} \odot \mathcal{H}) = \text{Mult}(\mathcal{H}).$$

Its Corollary 1.3 shows that the column–row property implies this equality.

Answer. Hartz, arXiv:2005.09614v1, Theorem 1.2 (page 3), proves:

Every normalized complete Pick space satisfies the column–row property with constant 1.

This is stronger than the requested existence of a finite constant and is optimal, since the definition requires the constant to be at least one. Hartz’s Theorem 1.4 (page 4) then explicitly combines Theorem 1.2 with Clouâtre–Hartz, Corollary 1.3, to obtain

$$\text{Mult}(\mathcal{H} \odot \mathcal{H}) = \text{Mult}(\mathcal{H}), \quad \|\varphi\|_{\text{Mult}(\mathcal{H} \odot \mathcal{H})} = \|\varphi\|_{\text{Mult}(\mathcal{H})}.$$

Thus both the finite-constant question and Question 1.1 are completely answered for normalized complete Pick spaces.

Provenance and scope. This is an explicit literature answer, not a new result of the run. The supporting paper cites the source paper and names its Corollary 1.3 in deriving Theorem 1.4. No portion of the normalized complete Pick case remains open. Questions classifying more general reproducing-kernel Hilbert spaces or arbitrary operator spaces with the column–row property lie outside the source question.

References

- [1] R. Clouâtre and M. Hartz, *Multipliers and operator space structure of weak product spaces*, Analysis & PDE 15 (2022), arXiv:1909.12883.
- [2] M. Hartz, *Every complete Pick space satisfies the column–row property*, Acta Mathematica 231 (2023), arXiv:2005.09614.